

PART A – OBJECTIVE QUESTIONS
[BAHAGIAN A – SOALAN OBJEKTIF]

[15 marks]
[15 markah]

Part A consists of 15 objective questions. Choose the best answer, and write your answer in the answer sheet provided in **page 2**. Each question carries 1 mark.

*[Bahagian A mengandungi 15 soalan objektif. Pilih jawapan yang paling tepat, dan jawab pada kertas jawapan yang disediakan pada **mukasurat 2**. Setiap soalan bernilai 1 markah.]*

1. Which of the following is the term used to describe the internal representation of an object that is hidden from the view outside of the object's definition?
[Manakah antara berikut merupakan istilah yang digunakan untuk menerangkan perwakilan dalaman sesuatu objek yang disembunyikan daripada pandangan luar bagi penakrifan objek?]

A. Encapsulation [Encapsulasi]	C. Expandable [Boleh Kembang]
B. Polymorphism [Polimorfisme]	D. Inheritance [Pewarisan]

2. Which of the following has the ability to focus on the important features of an object when trying to handle large amount of information.
[Manakah antara berikut berkemampuan untuk memfokus kepada ciri-ciri penting objek apabila mengendalikan maklumat yang banyak?]

A. Inheritance [Pewarisan]	C. Polymorphism [Polimorfisme]
B. Abstraction [Pengabstrakan]	D. Generality [Umum]

3. Which of the following is the program that interprets the Java bytecode?
[Manakah antara berikut merupakan atur cara yang menterjemah kod bait Java?]

A. Java API
B. Java debugger
C. Java virtual machine
D. Java compiler

4. Which of the following are valid names for Java variables?
[Manakah antara berikut merupakan nama yang sah bagi pembolehubah Java?]

- i. &ersand
- ii. 1Malaysia
- iii. One_Malaysia
- iv. A To Z
- v. Double_To_Float

- A. i, ii, and iii
- B. i, ii, iii, and v
- c. i, iii, and v
- d. i, ii, iii, and iv.

5. Which of the following applies overloading Java?

[Manakah antara berikut merupakan melaksanakan konsep limpahan dalam Java?]

- A. Method and attributes *[Kaedah dan atribut]*
- B. Method and constructor *[Kaedah dan Konstruktor]*
- C. Constructor and attributes *[Konstruktor dan atribut]*
- D. None of the above. *[Tiada yang di atas]*

6. Assume the definition of **myclass** is as in Figure 1, which of the following is correct to instantiate **myclass**?

*[Andaikan pentakrifan **myclass** seperti dalam Rajah 1, manakah yang berikut adalah BENAR untuk mentikakan **myclass**?]*

```
public class MyClass{
public MyClass() { /*code*/ }
// more code... }
```

Figure 1

- A. **MyClass mc = new MyClass() ;**
- B. **MyClass mc = MyClass() ;**
- C. **MyClass mc = new MyClass(10) ;**
- D. **MyClass mc = new MyClass;**

Given the program in Figure 2, answer question 7,8, and 9.

[Diberi atur cara pada Rajah 2, jawab soalan 7, 8 dan 9.]

```
public class Test {  
    public static void main(String[] args) {  
  
        double d = 0;  
        for ( d = 0; d < 2;) {  
  
            int i = 10;  
            int j = 100;  
            int k = ++j + j * 5 + i++;  
  
            System.out.println("i is " + i);  
            System.out.println("j is " + j);  
            System.out.println("k is " + k);  
  
            d += 0.5; }  
        }  
    }  
}
```

Figure 2

7. What is the value of the variable **k** after the first iteration of the loop?

[Apakah nilai pembolehubah **k** selepas pusingan pertama gelung?]

- A. 617
- B. 616
- C. 612
- D. 610

8. How many times will the **do** loop executes?

[Berapa kalikah gelung **do** diulang?]

- A. 2
- B. 3
- C. 4
- D. 5

9. What is the final value of **d** after the loop ends ?
[Apakah nilai terakhir bagi **d** selepas tamat gelung?]

- A. 0.5
- B. 1.0
- C. 1.5
- D. 2.0

10. Analyze the following code in Figure 3. What will happen if the code is compiled and/or executed?
[Analisa atur cara pada Rajah 3. Apakah yang akan berlaku sekiranya atur cara tersebut dikompil dan/ atau dilaksanakan?]

```
public class Test {  
    public static void main(String[] args) {  
        int n = 2;  
        xMethod(n);  
        System.out.println("n is " + n);  
    }  
  
    void xMethod(int n) {  
        n++;  
    }  
}
```

Figure 3

- A. The code has a syntax error because **xMethod** does not return a value.
[Atur cara mempunyai ralat sintaks kerana **xMethod** tidak mengembalikan nilai]
 - B. The code has syntax error because **xMethod** is not declared **static**.
[Atur cara mempunyai ralat sintaks kerana **xMethod** tidak diisytiharkan **static**]
 - C. The code prints **n is 3**.
[Atur cara akan mencetak **n is 3**]
 - D. The code prints **n is 2**.
[Atur cara akan mencetak **n is 2**]
11. Given the following statement which declares an enumerated data type for a variable named **menu**, which is the correct initialization with **NOODLE** constant?

[Diberi pernyataan berikut yang mengisytiharkan jenis data pengangkaan bagi pembolehubah bernama **menu**, manakah di antara berikut pemberi nilai awal yang betul kepada pemalar **NOODLE**?]

```
enum Food { BURGER , PIZZA , NOODLE , RICE }
```

- A. `enum menu = Food.NOODLE;`
- B. `Food menu = Food.NOODLE;`
- C. `Food menu.NOODLE;`
- D. `Food menu = menu.NOODLE;`

12. Which statement is **NOT CORRECT** for Java package?

[Pernyataan manakah yang **TIDAK BENAR** bagi pakej Java?]

- A. Java Virtual Machine (JVM) knows where the packages are located by setting the CLASSPATH environment variable.
[JVM tahu di mana kedudukan pakej melalui pembolehubah persekitaran CLASSPATH.]
- B. Java uses file system directories to maintain packages.
[Java menggunakan direktori system fail untuk menyenggara pakej.]
- C. `package myfavcar;` is the correct statement to declare a package.
[`package myfavcar;` adalah pernyataan yang betul untuk mengisytiharkan pakej.]
- D. `import myfavcar { BENTLEY, BMW, FERRARI };` is the correct statement to import a package.
[`import myfavcar { BENTLEY, BMW, FERRARI };` adalah pernyataan yang betul untuk mengimpor pakej.]

13. Given the following source codes in Figure 4, what is the expected sequence of numbers result from the code execution?

[Diberi kod sumber berikut dalam Rajah 4, apakah jangkakan turutan nombor yang akan dihasilkan apabila kod dilaksanakan?]

1	<code>int [] myArray = {10,20,30,40,50,60,70,80,90,100};</code>
2	<code>for (int i = 1; i < 7; i = i+2)</code> <code>System.out.print (myArray[i]);</code>

3	
3	

Figure 4

- A. 10 30 50 60
- B. 20 40 60 80
- C. 20 40 60
- D. 10 30 50

14. Which is the correct output for the Java codes in Figure 5?
 [Manakah output yang betul bagi kod Java pada Rajah 5?]

1	<code>int [][] table = new int[3][];</code>
2	<code>table[0] = new int[2];</code>
3	<code>table[1] = new int[5];</code>
4	<code>table[2] = new int[4];</code>
5	<code>System.out.print(table.length+" ");</code>
6	<code>System.out.println(table[1].length);</code>

Figure 5

- A 3 5
- .
- B 2 5
- .
- C 5 4
- .
- D 5 3
- .

15. Assume that the declaration of **CapitalCity** class is as in Figure 6.
 [Andaikan pengisytiharan kelas **CapitalCity** adalah seperti dalam Rajah 6.]

1	class CapitalCity {
2	private String country;
3	private String capCity;
4	
5	public CapitalCity(String cnt, String cty) {
6	country = cnt;
7	capCity = cty;
8	}
9	
10	public String getCapital() {
11	return capCity;
12	}
13	}

Figure 6

Which of the following is the correct statement that uses **enhanced for-loop** to display all the country name of the following array of objects?

*[Manakah antara berikut adalah pernyataan yang betul menggunakan **gelung enhanced-for** untuk memaparkan semua nama negara bagi tatasusunan objek berikut?]*

```
CapitalCity [] cc = {new CapitalCity{"Malaysia", "Kuala Lumpur"},
                     new CapitalCity{"Indonesis", "Jakarta"},
                     new CapitalCity{"Japan", "Tokyo"}};
```

- A for (int i=0; i<cc.length; i++) {
 System.out.println (cc[i].getCapital());
 }

- B for (String val : cc) {
 System.out.println (cc.getCapital());
 }

- C for (CapitalCity val : cc) {
 System.out.println (cc.getCapital());
 }

- D for (CapitalCity val : cc) {
 System.out.println (val.getCapital());
 }

PART B - STRUCTURED QUESTIONS
BAHAGIAN B – SOALAN STRUKTUR

[55 marks]
[55 markah]

Part B consists of 6 structured questions. Answer all questions in the space provided. The marks for each part of the question is as indicated.

[Bahagian B mengandungi 6 soalan struktur. Jawab semua soalan pada ruang yang disediakan. Markah setiap soalan adalah seperti yang dinyatakan.]

Question 1

[7 marks]

Given Program 1.1 as follows. Write the steps for running a Java program from command prompt.

[Diberi Program 1.1 berikut. Tuliskan langkah-langkah untuk melaksanakan aturcara Java melalui prompt arahan.]

```
// Program 1.1
public class myFirstTest {
    public static void main(String[] args) {
        System.out.println("GOOD LUCK!");
    }
}
```

- a) Save the file with the name : *[Simpan fail ini dengan nama :]*

- b) Below is some content of a file that we need to prepare in order to run a java program.

What is the name of this file ? _____

Please write the missing command in the file.

[Berikut adalah sebahagian kandungan fail yang perlu disediakan untuk melaksanakan aturcara java. Apakah nama fail tersebut? Tuliskan arahan yang tertinggal di dalam fail ini.]

```
set _____=C:\Program Files\Java\jdk1.6.0\bin
set _____=C:\JavaProg
```

- c) Write the command to compile the program : _____

[Tuliskan arahan untuk mengkompil aturcara tersebut.]

- d) What file will be produced after you compile the program:

[Apakah fail yang akan terhasil setelah aturcara dikompil?]

e) Write the command to run the program

[Tuliskan arahan untuk melaksanakan aturcara ini.] :

Question 2

[9 marks]

The following program has 5 errors. Circle the error and rewrite the program with the correct statements.

[Atur cara berikut mempunyai 5 ralat, bulatkan ralat tersebut dan tulis semula aturcara dengan pernyataan yang betul.]

```
import javax.swing.JOptionPane;
public class Test {
    public void main(string[] args ) {
        int count = 0;
        int num = 100.98;
        float num1 = 10.05f;
        String name;
        Scanner inp = new Scanner(System.in);
        do {
            System.out.print("Enter a letter:");
            name = inp.nextString();
            System.out.println("num is " + num + "Num1 is " + num1);
            System.out.println("the name is " + name);
            count++;
        } while (count < 3);
    }
}
```

Question 4

[7 marks]

Write 3 overloading static method, named **min** that able to find the minimum value from parameters of different data types as invoked in the following program.

[Tulis tiga kaedah static limpahan bernama **min** yang boleh mencari nilai minimum daripada parameter yang berbeza dari segi jenis data seperti yang dipanggil dalam atur cara berikut.]

1	public class MethodOverLoading
2	{
3	public static void main (String [] args)
4	{
5	

6	System.out.println ("The minimum between 120 and 135 is
7	" + min (120, 135));
8	System.out.println ("The minimum between 141.2 and
9	135.7 is " + min (141.2, 135.7));
10	System.out.println ("The minimum between 110.7, 107.3
11	and 115.3 is " + min (110.7, 107.3, 115.3));
12	}
	}

Question 5

[9 marks]

Given a program below, answer the following question:

[Diberi program di bawah, jawab soalan-soalan berikut:]

1	enum Day {
2	SUNDAY ("AHAD"), MONDAY ("ISNIN"), TUESDAY ("SELASA"),
3	WEDNESDAY ("RABU"), THURSDAY ("KHAMIS"), FRIDAY ("JUMAAT"),
4	SATURDAY ("SABTU");
5	
6	private String shortName;
7	private Day (String shortName) {
8	this.shortName= shortName;
9	}
10	public String getShortName() {
11	return shortName;
12	}
13	}

- i. Write a statement that declare a variable of enumerated data type, named **testDay** and initialized with enum value WEDNESDAY.

[Tuliskan satu pernyataan yang mengishtiharkan satu pembolehubah berjenis enumerated bernama **testDay** dan diberi nilai awal kepada nilai enum WEDNESDAY]

[2 marks]

- ii. What will be the output of the following statement?

[Apakah output bagi pernyataan di bawah?]

```
System.out.println("The ordinal value for " + Day.FRIDAY + "
is " + Day.FRIDAY.ordinal());
```

[2 marks]

- iii. Write the correct statement to invoke the method named **getShortName()** of the constant name **SATURDAY**.

*[Tuliskan satu pernyataan untuk memanggil metod bernama **getShortName()** bagi pemalar bernama SATURDAY.]*

[2 marks]

- iv. Write an if statement that will print message “I have test today” if testDay is WEDNESDAY” and will print message “No Test Today.” for any other value.

*[Tuliskan satu pernyataan if yang akan mencetak mesej “I have test today” jika **testDay** ialah WEDNESDAY dan mencetak mesej “No Test Today” jika **testDay** bukan WEDNESDAY]*

[3 marks]

PART C – PROGRAMMING QUESTION
BAHAGIAN C – SOALAN ATUR CARA

[30 marks]
[30 markah]

Part C consists of 1 question. Answer the question in the space provided. The marks for each part of the question is as indicated.

[Bahagian C mengandungi 1 soalan. Jawab semua soalan pada ruang yang disediakan. Markah setiap soalan adalah seperti yang dinyatakan.]

The Table below gives the descriptions of attributes and method for class **Employee**.

*[Jadual di bawah member penerangan mengenai atribut dan kaedah bagi kelas **Employee**.]*

Member variables	
empID : string	is of type <code>String</code> that hold the employee ID
empName : string	is of type <code>String</code> that hold the employee name
totalSalary : double	is of type <code>double</code> that hold the total salary for the whole year
tax : double	is of type <code>double</code> that hold the tax that the employee has to pay
Member Functions	
readValue()	The function will read value for <code>empID</code> , <code>empName</code> and <code>totalSalary</code> from the keyboard.
void display()	The function print the employee information and the tax that he has to pay.
	The accessor method for all attributes
	The mutator method for all attributes

Answer the questions given in this section.

[Jawab soalan pada bahagian ini]

1. Draw class diagram for class **Employee** as described in the table.

*[Lukiskan rajah kelas bagi kelas **Employee** seperti yang ditunjukkan dalam jadual.]*

[3 marks]

2. Write a complete Java Program that consists of the following:

[Tuliskan atur cara Java yang lengkap yang mempunyai perkara berikut:]

- a. Declare **Employee** class with the private member variables and the source codes for the member functions as shown in the table.

[Isytiharkan kelas **Employee** dengan ahli pembolehubah private dan kod sumber bagi ahli kaedah seperti yang ditunjukkan dalam jadual.]

[10 marks]

- b. Declare **EmployeeTest** class that has the following method declaration and codes.

[Isytiharkan kelas **EmployeeTest** yang mempunyai pengisytiharan kaedah dan kod sumber.]

calcTax (Employee)

The function will calculate the income tax for each employee based on these criteria.

1. Total salary less than RM30,000.00 - no tax will be charged
2. Total salary 30,000 – 60,000 - The first 30,000 will be no tax, and the rest will be charged 5% of the balance
3. Total salary greater than 60,000 – the first 30,000 will be no tax, the next 30,000 will be charged 5% and the rest will be charged 8%.

[Fungsi ini akan mengira cukai pendapatan setiap pekerja berdasarkan kriteria berikut:]

1. Jumlah pendapatan kurang dari RM30,000.00 – tiada cukai dikenakan.
2. Jumlah pendapatan sebanyak RM30,000 – RM60,000 - RM30,000 yang pertama tidak dikenakan cukai, baki selebihnya dikenakan cukai sebanyak 5%.
3. Jumlah pendapatan melebihi RM60,000 - RM30,000 yang pertama tidak dikenakan cukai, RM30,000 yang kedua dikenakan cukai sebanyak 5% dan baki selebihnya dikenakan cukai sebanyak 8%.]

[8 marks]

highestSalary (Employee emp[])

The function will find the highest salary among the employee and print the employee information.

[Fungsi ini akan mencari pekerja dengan gaji tertinggi dan mencetak maklumat pekerja tersebut.]

[4 marks]

Main method

- a. Creates an array of 10 Employee objects, named **emp**. Use enhanced loop to read values for the 10 employee.
- b. Use loop to calculate tax for each employee in the array.
- c. Display all employee information and the tax that they have to pay.
- d. Find the employee with the highest salary and print his information.

[5 marks]